

The Fuzzy Subjective Structure Model (FSSM)

Evaluating Causal Claims for Non-Deterministic Inference on Subjective Constructs

Marek Deja

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Jagiellonian University

The Generalization Impasse in LIS

- Library and Information Science has built rich theoretical frameworks, but empirical studies remain bound by methods oriented toward population-wide generalization.
- A typical information behaviour research project in context requires its own specific design and methodological development.
- The full potential of in-context research is not in chasing sweeping averages, but in supporting decision-making based on local evidence.
- We need dedicated designs that evaluate context-specific configurations rather than default generalization.

Subjective Structures & Effects of Causes

- Subjectivity is the very area where the effect of causes is located.
- FSSM captures how the effects of a cause—whether adjacent (proximal interaction) or coercive (structural policies, culture, and norms)—reside within the subjective structure of human perception rather than acting as deterministic drivers.
- Environmental and institutional factors do not directly force behaviour; their impact is filtered through how individuals subjectively perceive and experience affordances and constraints.

Beyond the “Average” Human

- Traditional regression models rely on mean estimates—a vague abstraction of a “perfect human being” that none of us is.
- Such approaches discard the individual autonomy of thoughts and actions in context.
- FSSM shifts focus to subjectivity mapping, assessing:
 - Typicality: regularities that hold for nearly all cases in a group.
 - Necessity: non-compensatory system bottlenecks—critical gatekeepers.
 - Probability: individual likelihood of behaving as an effect of context causes.
- We examine how far or close individuals are from modeled desirable thresholds rather than computing aggregate averages.

Arrow-Reversal & Context-First Skeleton



- Theory dictates: context shapes perception, which drives behaviour.
- Many models regress culture on behaviour, reversing the causal order. Compliance with open data policies is not an “effect” producing culture; institutional norms enable action.
- FSSM enforces the skeleton: Management and Culture as upstream enabling conditions; Self-Efficacy and Empowerment as intermediate perceptions; Collaboration as the downstream behaviour.

The Four Causal Claims of FSSM

1. Typical Sufficiency (α): “If $X \geq t_X$, then $Y \geq t_Y$ for most cases.” Policies usually suffice to produce the behaviour; rare exceptions are allowed.
2. Typical Necessity (ε): “If Y is high, X is rarely below its bottleneck.”
Non-compensatory constraint—a gatekeeper whose absence blocks success.
3. Probabilistic Sufficiency (Δ / Uplift): Does raising X above the threshold increase the probability of high Y ? Computed on an IPW-balanced pseudo-population.
4. Probabilistic Necessity (β): “When X is below its bottleneck, high Y is unlikely.” Converts the constraint into a counterfactual risk statement.

Four Causal Claims: Mathematical Form

$$\alpha = \frac{\sum_i w_i \cdot \mathbf{1}(x_i \geq t_X) \cdot \mathbf{1}(y_i \geq t_Y)}{\sum_i w_i \cdot \mathbf{1}(x_i \geq t_X)} \geq 0.85$$

$$\varepsilon = \frac{\sum_i w_i \cdot \mathbf{1}(y_i \geq t_Y) \cdot \mathbf{1}(x_i < t_X)}{\sum_i w_i \cdot \mathbf{1}(y_i \geq t_Y)} \leq 0.15$$

$$\Delta = \Pr^*(Y \geq t_Y \mid X \geq t_X) - \Pr^*(Y \geq t_Y \mid X < t_X)$$

$$\beta = \Pr^*(Y \geq t_Y \mid X < t_X) \leq 0.20$$

t_X (NCA-derived bottleneck) and t_Y (desired outcome level) anchor all four claims to the same empirical scale. $\Pr^*$ denotes probability on the IPW-balanced pseudo-population; w_i are normalized case weights.

Dual Logic: Latent vs. Composite Route

- Route A (Latent Variable Route — PLS-SEM + NCA): Extends latent structural models. Achieves population generalization via structural paths while exploring necessity bottlenecks case by case. A sound ground for causal generalization with reversal to case-wise interpretation.
- Route B (Raw Composite Route — NCA + Case Weighting): Bypasses latent modelling by using raw composites. Enables a case-level “finding people” approach: NCA identifies the threshold t_X for a specific desired t_Y , and every respondent is located on the resulting membership scale.
- Route B allows researchers to study teams—even small ones—to diagnose their specific information needs and behaviours based on local evidence, without demanding full latent measurement.

Double Subjectivity: Entropy Weighting & IPW

FSSM addresses subjectivity at two levels:

1. Measurement subjectivity — respondents read items through disciplinary lenses. Entropy weighting up-weights discriminating indicators, down-weights noise:

$$w_j^{\text{EW}} \propto H(j) = - \sum_k p_{jk} \log p_{jk}$$

2. Confounding — group differences may reflect discipline or seniority, not the construct. IPW re-balances the sample:

$$w_i^{\text{IPW}} = \frac{D_i}{\hat{e}(c_i)} + \frac{1 - D_i}{1 - \hat{e}(c_i)}, \quad \hat{e}(c_i) = \Pr(D_i = 1 \mid c_i)$$

- Together, these two layers move FSSM from association toward principled intervention contrasts under explicitly stated identification assumptions.

Fuzzy Concept & Subjectivity Mapping

- Continuous latent and composite scores are mapped onto a fuzzy membership interval $[0, 1]$ relative to empirical thresholds and bottlenecks.
- $\mu_S(x)$: degree to which an individual case meets the necessary context bottleneck—how close a case is to crossing the enabling threshold.
- $\mu_Y(y)$: degree of outcome achievement relative to the desired target t_Y .
- Graded membership honours individual autonomy: it shows precisely who clears the threshold, who falls short, and who represents a critical exception—without binary cutoffs.

Fuzzy Membership: Mathematical Form

Membership in the necessary bottleneck for condition X:

$$\mu_S(x_i) = \begin{cases} 0 & x_i = 0 \\ \frac{x_i}{t_X} & 0 < x_i < t_X \\ 1 & x_i \geq t_X \end{cases} \quad \mu_Y(y_i) = \min\left(1, \frac{y_i}{t_Y}\right)$$

The pair $(\mu_S(x_i), \mu_Y(y_i))$ locates every respondent on a self-treatment scale. The institutional environment is an ongoing exogenous context continuously shaping each individual's endogenous subjective structure. Threshold-seeking replaces point-treatment logic: not “did an intervention occur?” but “how far is this person from the enabling threshold?” In FSSM, treatment is replaced with threshold-seeking as an individual's ongoing reaction to persistent environmental conditions.

Thank You for Your Attention

Marek Deja

Assistant Professor, Jagiellonian University

`marek.deja@uj.edu.pl`

FSSM Project & Updates on OSF:

`osf.io/b8maq/`